

ROUNSAVILLE TECHNOLOGIES

R-COREX INTELLIGENCE ARCHITECTURE

CODE-TO-CONCEPT MAPPING RECORD

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Document ID: RT-IP-RCOREX-MAPPING-002 **Version:** 1.0.0 **Status:** Supporting Evidence Record — companion to RT-IP-RCOREX-MASTER-001 **Author:** Joseph Michael Rounsaville / Rounsaville Technologies (compiled by session record) **Organization:** Rounsaville Technologies **Record Date:** July 23, 2026 **Document Classification:** Proprietary / Confidential — Internal Master Record **Supersedes:** Section 16 ("Code-to-Concept Mapping") of RT-IP-RCOREX-MASTER-001_v1.0.0, which used placeholder paths (e.g. /core/, /bus/) not yet tied to any repository.

1. Purpose and Method

RT-IP-RCOREX-MASTER-001 v1.0.0 described the R-CoreX Intelligence Architecture in the abstract, with a Section 16 mapping table of placeholder paths pending future implementation.

This record replaces that placeholder with a mapping against the **actual repositories** in the Rounsaville Technologies GitHub account, verified by direct inspection of each repository's real, current content as of the record date. Every path cited below was read from the live repository — none are inferred or assumed.

Method: each of the three repositories in scope was cloned and inspected directly (directory structure, README files, source files, docstrings, and commit history). Where a concept has no real counterpart in a repository, this record says so explicitly rather than forcing a mapping.

2. Repositories Surveyed

Repository	Branch / Commit Inspected	Visibility	Summary
presencefieldowner-glitch/RealityStateOS	main @ 8e41ae2	Private	14-module (000–999) "verifiable, agent-native operating system." Architecture (000), Kernel (002), and Foundation (003) modules are built with passing tests. Modules 001, 004–012, 999 are named and ordered in the repo's own Build Order but not yet scaffolded.

Repository	Branch / Commit Inspected	Visibility	Summary
presencefieldowner-glitch/rounsaville-ai-music-generator	— (no commits)	Private	Empty repository. git clone succeeds but returns "you appear to have cloned an empty repository." No branches, no files, nothing to map.
presencefieldowner-glitch/presence-resonance	main @ a3f9727	Public	A genealogy/family-history platform ("reconstructs family trees... maps them across time and geography"), by the same author. pnpm monorepo scaffold; every one of its 11 workspace packages contains only a README and an empty export <code>{}</code> ; — no implemented logic anywhere.

3. Central Finding: R-CoreX Is RealityStateOS, Described Abstractly

R-CoreX is not a separate codebase sitting apart from these products. Reading RealityStateOS/README.md's own module map alongside the dossier makes the relationship unmistakable:

Dossier concept (abstract)	RealityStateOS module (concrete, self-described)
R-CoreX (parent architecture)	The whole 14-module OS
Verified Bus	004_VERIFIED_BUS — "Event backbone: broker, consensus, verification, telemetry" (RealityStateOS's own words, independent of this dossier)

Dossier concept (abstract)	RealityStateOS module (concrete, self-described)
Nine-Phase Interpretation Pipeline	009_INTELLIGENCE ("Planning, reasoning, learning, agent runtime, workflows") at the module level; concretely prototyped today inside 002_KERNEL/LakeTiticaca as a 5-step cognition loop
SRI (sensory layer)	008_REALITY_TAPE — "Sensor ingestion, state reconstruction, timeline replay, simulation"
NHI (interface layer)	011_INTERFACE — "REST/GraphQL/WebSocket/CLI/SDK/Mobile/Desktop/Web/XR surfaces"
Telemetry Fabric — Identity	007_PRESENCE — "Identity, auth, presence scoring, trust engine"
Telemetry Fabric — Provenance / Knowledge Retrieval	005_KNOWLEDGE — "Knowledge repository, semantic search, provenance, versioning"
Relationship Mapping	006_REALITY_GRAPH — "nodes, edges, temporal/spatial/trust dimensions"
Event Schemas / Verification	003_FOUNDATION — core types, crypto, serialization
Audit Trails / Verification enforcement	999_OPERATIONS — "deployment, monitoring, security, compliance, CI/CD"

This means the dossier's architecture and RealityStateOS's architecture are, on the evidence, **one and the same system described at two different altitudes** — the dossier is the conceptual/IP framing, RealityStateOS is the product build-out. Treat future dossier revisions and RealityStateOS's own 000_ARCHITECTURE RFCs as a single source of truth, not two independent designs.

4. The Real Cognition Loop vs. the Dossier's Nine Phases

RealityStateOS/002_KERNEL/LakeTiticaca/ ("Kernel-level AI," Python, single-node prototype, has passing tests) implements a cognition loop documented in its own README.md:

```
Observe -> Interpret -> Align -> Reason -> Evolve
```

This is the real, working ancestor of the dossier's nine-phase pipeline — but it is **five steps, not nine**, in actual code:

- `observation.py` — Phase 1 (Observation): "append raw inputs, uninterpreted."
- `interpretation.py` — Phase 2 (Interpretation): "rule-based tagging into structured Findings."
- `alignment.py` — Phase 3 (Alignment): governance/policy gate — "the Kernel AI is not allowed to act outside declared bounds."
- `reasoning.py` + `inference.py` — Phase 4 (Reasoning), absorbing Phases 5–7:
- `retrieval.py` unifies `semantic_index.py`, `vector_store.py`, and `knowledge_graph.py` lookups — this is Phase 5 (Knowledge Retrieval), implemented as a helper called from Reason, not a standalone loop step.

- `relationships.py` — Phase 6 (Relationship Mapping), a Python-side mirror of 003_FOUNDATION/EntityModel's typed relationship semantics.
- `inference.py` — Phase 7 (Inference), "minimal forward-chaining rule engine."
- **Phase 8 (Response Construction) does not exist as a loop step.** Output happens through separate interface-surface modules — `api.py` (transport-agnostic command registry), with `rest.py`, `grpc.py`, `websocket.py` as thin gateways over it, and `cli.py` / `dashboard.py` as developer-facing tools. `api.py`'s own docstring notes these are prototype-stage; binding real servers is explicitly deferred to the planned 011_INTERFACE module.
- `evolution.py` — Phase 9 (Evolution): "records how LakeTiticaca's own knowledge/config changed," layered on `version_control.py`.

Implication for the IP record: the dossier's claim to a discrete "nine-phase" method should be understood, on current evidence, as a five-phase real implementation with three phases collapsed into one and one phase (Response Construction) not modeled as a loop step at all. Future patent or trade-secret claims specific to a nine-step sequence are not yet supported by the reduction-to-practice on file; the five-step loop is what has actually been built and tested.

5. Full Concept-to-Code Table

Concept	Real Path(s)	Repository	Status
R-CoreX (parent)	/ (whole repo, README.md module map)	RealityStateOS	Host product, scaffolding phase
Observation	002_KERNEL/LakeTiticaca/laketiticaca/observation.py	RealityStateOS	Implemented (prototype)
Interpretation	002_KERNEL/LakeTiticaca/laketiticaca/interpretation.py	RealityStateOS	Implemented (prototype)
Alignment	002_KERNEL/LakeTiticaca/laketiticaca/alignment.py	RealityStateOS	Implemented (prototype)
Reasoning	002_KERNEL/LakeTiticaca/laketiticaca/reasoning.py	RealityStateOS	Implemented (prototype)

Concept	Real Path(s)	Repository	Status
Knowledge Retrieval	002_KERNEL/LakeT iticaca/laketiti caca/retrieval.p y, semantic_index .py, vector_store.py, knowledge_graph. py	RealityStateOS	Implemented (prototype); collapsed into Reason
Relationship Mapping	002_KERNEL/LakeT iticaca/laketiti caca/relationshi ps.py; 003_FOUNDA TION/EntityModel /src/entityModel .ts	RealityStateOS	Implemented (prototype); collapsed into Reason
Inference	002_KERNEL/LakeT iticaca/laketiti caca/inference.p y	RealityStateOS	Implemented (prototype); collapsed into Reason
Response Construction	002_KERNEL/LakeT iticaca/laketiti caca/api.py, rest.py, grpc.py, websocket.py, cli.py, dashboard.py	RealityStateOS	Implemented (prototype); not a loop step
Evolution	002_KERNEL/LakeT iticaca/laketiti caca/evolution.p y	RealityStateOS	Implemented (prototype)
Verified Bus (prototype)	002_KERNEL/LakeT iticaca/laketiti caca/event_bus.p y	RealityStateOS	Implemented (prototype)
Verified Bus (full module)	004_VERIFIED_BUS /	RealityStateOS	Named in module map; not yet scaffolded
IHI	—	RealityStateOS	No dedicated module; distributed across retrieval/knowledge modules

Concept	Real Path(s)	Repository	Status
RHI	reasoning.py, inference.py	RealityStateOS	Implemented (prototype)
NHI (prototype)	api.py, rest.py, grpc.py, websocket.py, cli.py, dashboard.py	RealityStateOS	Implemented (prototype)
NHI (full module)	011_INTERFACE/	RealityStateOS	Named in module map; not yet scaffolded
SRI (minimal)	observation.py	RealityStateOS	Minimal — raw intake only
SRI (full module)	008_REALITY_TAPE /	RealityStateOS	Named in module map; not yet scaffolded
Event Schemas	003_FOUNDATION/C oreTypes/src/eve nt.ts	RealityStateOS	Implemented — hash-chained, signed Event<T> envelope (RFC-0002 §5)
Identity (prototype)	authentication.p y, authorization.py	RealityStateOS	Implemented (prototype) — HMAC-signed tokens
Identity (full module)	007_PRESENCE/	RealityStateOS	Named in module map; not yet scaffolded
Provenance	provenance.py	RealityStateOS	Implemented
Verification	003_FOUNDATION/C oreTypes/src/cry pto.ts, event.ts (verifyEvent, verifyChain); security.py, encryption.py	RealityStateOS	Implemented
Audit Trails	version_control. py	RealityStateOS	Implemented — git-lite, append-only
Extension APIs	plugins.py	RealityStateOS	Implemented (prototype)

Concept	Real Path(s)	Repository	Status
(any concept)	—	rounsaville-ai-music-generator	No implementation — repository is empty
(any concept)	packages/resonance/, packages/ai-interpreter/ (naming echo only)	presence-resonance	No implementation — all packages are export {}; stubs

6. Per-Repository Notes

6.1 RealityStateOS — primary host

- OWNERSHIP.md states sole inventorship by Joseph Michael Rounsaville, first conception 2026-06-29, ownership statement dated 2026-07-07.
- Kernel modules are split between mechanical TypeScript (002_KERNEL/*, 8 of 14 planned sub-modules built, 33/33 tests passing) and the Python Kernel AI (LakeTiticaca), which the Kernel's own README.md says are "not bridged yet" — that bridging is explicitly called out as "a 004_VERIFIED_BUS-level concern for a later phase."
- Three Kernel sub-modules (Scheduler/, SecurityKernel/, PolicyEngine/) exist as empty TS folders kept only for directory continuity — their real implementation lives in LakeTiticaca (scheduler.py, security.py, alignment.py), a useful example of the repo's own internal code-to-concept redirection that this record mirrors in spirit.
- Three further Kernel concerns (Memory/, ProcessManager/, ThreadManager/) are spec-only (KERNEL_RFC.md) — deliberately not faked with code, since a sandboxed runtime cannot meaningfully prototype real OS-level memory/thread isolation.

6.2 rounsaville-ai-music-generator — empty

No content exists to map. This record intentionally leaves every R-CoreX concept unmapped against this repository rather than inferring intended structure. Re-run this mapping exercise once code is pushed.

6.3 presence-resonance — sibling product, not an R-CoreX asset

- Self-describes as a genealogy platform, unrelated in subject matter to R-CoreX/RealityStateOS.
- OWNERSHIP.md again names Joseph Michael Rounsaville as sole creator, first conception 2026-07-22.
- The packages/resonance and packages/ai-interpreter names are coincidental echoes of SRI ("Sensory/**Resonance**") and IHI ("**I**ntelligence") — both are empty export {}; stubs. Citing this repository as evidence of SRI/IHI reduction-to-practice would not be supportable; it should be tracked as a separate product in its own IP registry, sharing only common authorship with R-CoreX/RealityStateOS.

7. IP Registry Additions

Two new IP asset rows are proposed for RT-IP-RCOREX-MASTER-001 Section 13 (already applied in v1.1.0 of that document):

ID	Asset	Basis
RT-IP-0014	RealityStateOS — Host Product Implementation	14-module architecture; Architecture/Kernel/Foundation built and tested
RT-IP-0015	LakeTiticaca — Kernel AI Cognition Loop	Single-node prototype implementing the real ancestor of the Nine-Phase Pipeline

8. Limitations

This record establishes technical provenance and a documented mapping between conceptual architecture and real source code, as of the repository states inspected on July 23, 2026. It is not a substitute for legal advice, and it does not itself establish patent, trademark, or copyright rights — see Section 27 of RT-IP-RCOREX-MASTER-001 for the governing limitations, which apply equally here. Repository states change; this record should be re-run whenever a named-but-unbuilt module (e.g. 004_VERIFIED_BUS, 007_PRESENCE, 008_REALITY_TAPE, 011_INTERFACE) is scaffolded, or when rounsaville-ai-music-generator receives its first commit.

END OF MAPPING RECORD